

PALLAVI KOCHAR

in [pallavikochar7](#) | ✉ pallavikochar8@gmail.com | ☎ +1 (217) 418-2810 | 🌐 [Portfolio](#)

EDUCATION

University of Illinois Urbana-Champaign (UIUC) (Urbana-Champaign, IL) Jan 2025 - May 2026

- Master of Science in Finance (Specialization: Asset Management, Concentration: Data Analytics, Quantitative Finance)
- Research Assistant - DeFi Crypto Asset Pricing, Course Assistant - Financial Economics, MSF Program Ambassador
- Coursework: Machine Learning, Applied Portfolio Management, Advanced Financial Derivatives, Big Data Analytics

Indian Institute of Technology (IIT), Bombay (Mumbai, India) Jun 2018 - May 2022

- Bachelor of Technology (BTech) in Chemical Engineering (Minor: Industrial Engineering and Operations Research)
- Coursework: Numerical Analysis, Calculus, Economics, Probability and Stochastic Processes, Differential Equations

PROFESSIONAL EXPERIENCE

Alpha.ai | Data Product Engineer Intern (New York City, USA) Apr 2026 - May 2026

- Built a SHAP-based Explainable AI layer over model predictions, translating opaque outputs into per-feature attributions and optimizing the attribution pipeline to balance explanation fidelity with runtime for production deployment
- Designed a stock summary surfacing key headwinds and tailwinds for at-a-glance insight ahead of detailed model output

Kotak Securities Limited | Backend Engineer (Mumbai, India) Jun 2022 - Dec 2024

- Spearheaded end-to-end design of a fully automated SIP order auto-debit system for Smallcase baskets, building backend services in C#, SSIS, and SQL Server that process 10K+ monthly transactions with 99.9% real-time accuracy
- Built scalable AWS-based data pipelines integrating Salesforce CRM with internal enterprise systems via REST APIs, accelerating customer insight access and improving analytics workflow efficiency by 30%
- Developed and optimized 10+ backend micro-services and RESTful APIs for corporate actions, gain/loss processing, and data verification in the Portfolio Tracker, cutting IT workload by 50+ hours weekly
- Automated multi-stock price update workflows, saving 200+ IT hours monthly and improving intraday analytics reliability

KEY PROJECTS

Top-Down Approach Multi-Agent System (Course: Applied Portfolio Management) 🔄 May 2026 - Jun 2026

- Architected an 11-agent stock recommendation engine in Python and FastAPI for top-down macro analysis, sector-rotation scoring, and 3-scenario DCF valuation; 10-year backtest returned 20% CAGR and +5.2% alpha vs. benchmark
- Built a React and TypeScript dashboard visualizing an 8-stage economy-to-stock investment funnel with sector heatmaps and confidence attribution, backed by REST APIs and 10+ pytest integrity checks

Satellite-anchored Multi-Agent Trading System (Course: Quantamental Investment) 🔄 Apr 2026 - May 2026

- Engineered a multi-agent equity-trading system for Permian E&P names enforcing a deterministic Python numerical core with GPT-4o-mini agents confined to qualitative roles, eliminating the LLM-calibration failure mode
- Backtested a satellite-anchored pre-earnings strategy across 10 oil & gas stocks; generated 18 trades with 83% hit rate, +13.5% return on \$1M, and 0.35 per-trade Sharpe, with pre-registered ablations validating signal source

Structured Product Valuation: Auto-Callable Securities (Course: Advanced Financial Derivatives) Apr 2026 - May 2026

- Priced a 2-year worst-of auto-callable note on AAPL/AMZN/GOOGL via Python Monte Carlo, simulating correlated GBM paths via Cholesky decomposition with Bloomberg-sourced ATM vols, OIS discounting, and continuous dividend drift
- Modeled path-dependent payoff (memory coupon, quarterly autocall, 50% barrier), priced at \$978/\$1,000 face, and quantified volatility and correlation sensitivities to isolate the embedded short worst-of put

Behavioral Analysis of Credit Card Advertising (Research Project) Jan 2026 - May 2026

- Engineered a Python pipeline to ingest and structure Competiscan data across thousands of credit-card ad creatives, then designed experiments isolating which product attributes most drive consumer selection and confidence
- Designed and ran controlled experiments measuring how feature salience and ad layout shift credit-card selection, quantifying which attributes most affect consumer preference

GARCH Volatility and VaR Modeling (Course: Financial Risk Management) Mar 2026 - Apr 2026

- Estimated a GARCH(1,1) volatility model on SPY returns via custom Maximum Likelihood in R, validating against tseries and fGarch implementations and forecasting 1-day and 21-day annualized volatility
- Computed 1-day VaR and Expected Shortfall (95%), benchmarking against historical-simulation and delta-normal methods

SKILLS AND AWARDS

- **Technical:** Python (Pandas, scikit-learn, TensorFlow, PyTorch), C#, SQL, R, REST APIs, ETL, CI/CD, Docker, AWS, Git
- **AI Tools:** Claude Code, Anthropic Cowork, GitHub Copilot, Cline, AntiGravity, multi-agent orchestration, Codex, MCP
- **Machine Learning:** Regression, Clustering, Time Series Forecasting, Neural Networks (LSTM, FinBERT), Explainable AI
- **Finance & Quant:** Derivative Pricing (Black-Scholes, Monte Carlo), Bloomberg, CapIQ, VaR, Portfolio Optimization
- **Certifications:** [Finance and Quantitative Modeling](#), [Machine Learning](#), [IBM Data Science](#), [LangChain](#)
- **Awards:** Advanced 3 designations at Kotak Securities in 2 years · IT Champion Award FY 2022-23 · Top 0.1% JEE-Mains